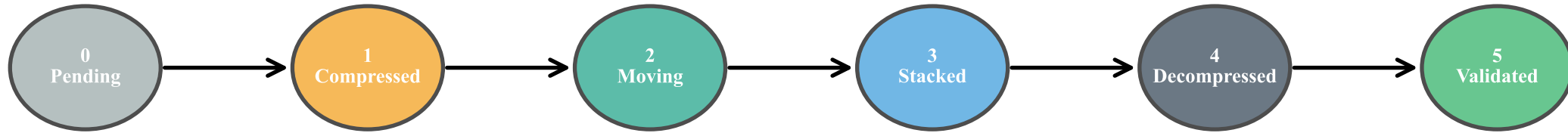


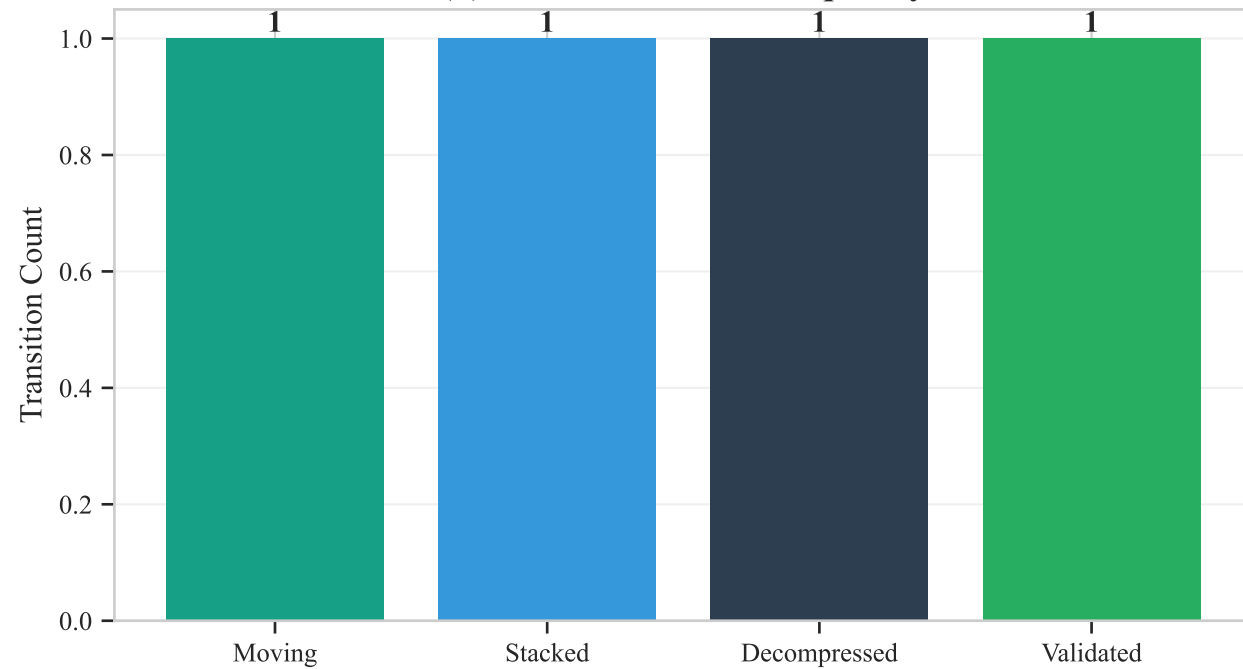
Figure 3: Six-State Transaction Lifecycle and State Transitions

(a) Transaction State Flow Diagram



State 0: Initial submission | States 1-4: Processing | State 5: Final validation

(b) State Transition Frequency



(c) Gas Cost per State Transition

